THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using the column addition algorithm, $194 + 355$ equals: 1

(a) 549 **(b)** 559 **(c)** 539 **(d)** 649

2. The divisor algorithm applied to 24 produces a list with how many entries? 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 10

3. The value of $38 \bmod 12$ (remainder on dividing 38 by 12) is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

4. The greatest of the common divisors of 30 and 22, found from their divisor lists, is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

5. An algorithm is best described as: 1

(a) a single formula
(b) a guess
(c) a diagram only
(d) a step-by-step procedure that solves a problem

6. Set total = 0. For i = 1, 2, …, 11: if i is even, add i to total. The final value of total is: 1

(a) 32 **(b)** 30 **(c)** 29 **(d)** 31

7. Euclid's observation used in the gcd algorithm is that gcd(m, n) equals: 1

(a) gcd(m × n, n) **(b)** m − n
(c) gcd(m, m + n) **(d)** gcd(n, m − n) for $m \geq n$

8. Adding 85 and 54 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 2 **(b)** 3 **(c)** 0 **(d)** 1

9. Using the column addition algorithm, $352 + 209$ equals: 1

(a) 561 **(b)** 551 **(c)** 571 **(d)** 661

10. The divisor algorithm applied to 12 produces a list with how many entries? 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 5

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Add 318 and 427 using the column algorithm. State in which columns a carry is produced. 2

12. Run the divisor algorithm for 14: check j = 1, 2, …, 14. Write the list of divisors and say how many numbers are checked. 2

13. Find $81 \bmod 12$ and $93 \bmod 12$. What do you notice? 2

14. Add 246 and 330 using the column algorithm. State in which columns a carry is produced. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find gcd(10, 30) by listing the divisors of both numbers. 3

16. Apply Euclid's subtraction algorithm to find gcd(39, 27). Write the sequence of pairs and count the subtractions. 3

17. Add 5840 and 4426 using the column algorithm. State in which columns a carry is produced. 3

OR

Find gcd(33, 18) by listing the divisors of both numbers. 3

18. Apply Euclid's subtraction algorithm to find gcd(42, 28). Write the sequence of pairs and count the subtractions. 3

OR

Add 3057 and 3038 using the column algorithm. State in which columns a carry is produced. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace this procedure: total = 0; for i = 1 to 14, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(50, 75) by listing the divisors of both numbers. 5

20. Trace this procedure: total = 0; for i = 1 to 10, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(66, 84) by listing the divisors of both numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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